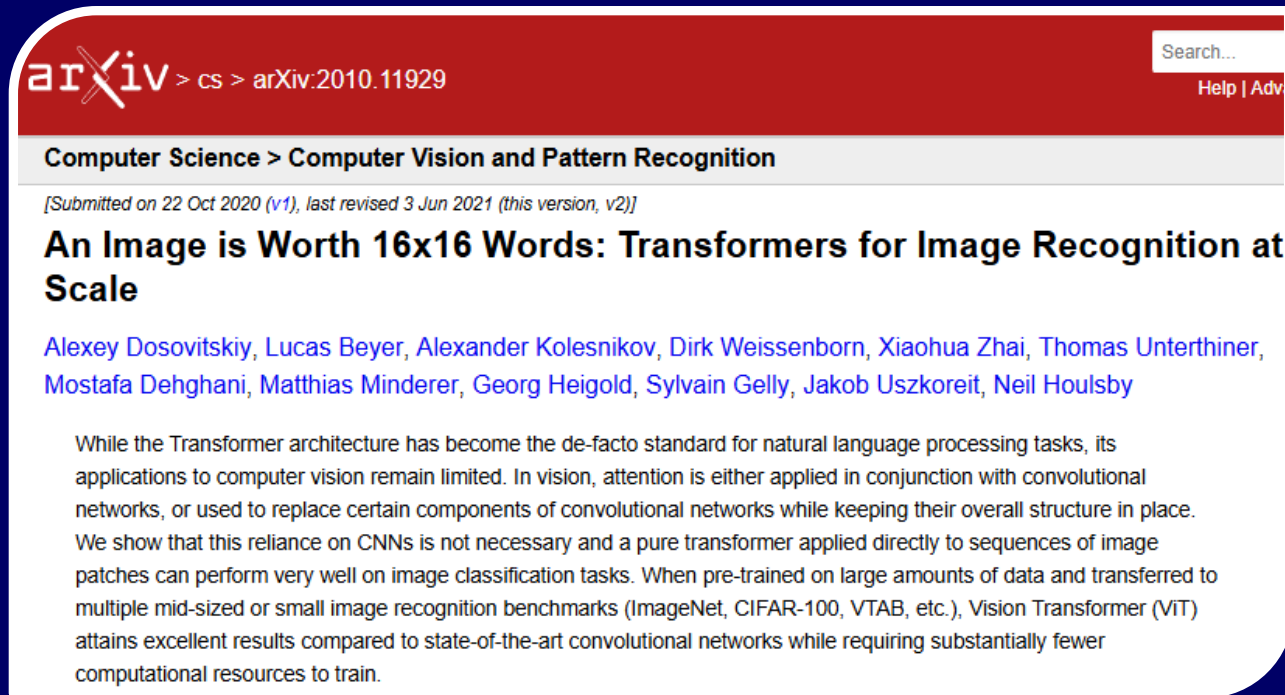




An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

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Introduction

Background

- In NLP, Transformers' computational efficiency and scalability, it has become possible to train models of unprecedented size, with over 100B parameters
- In vision, convolutional architectures remain dominant.
- There are some works to combine CNN with self-attention and replace whole architecture to self-attention.
- But hardware it does not match with modern HW accelerators.
- Therefore, in large-scale image recognition, classic ResNet like architectures are still SOTA.
- Inspired by the Transformer scaling successes in NLP, we experiment with applying a standard Transformer directly to images, with the fewest possible modifications.

Related work

Self-attention

- Transformer based NLP models become the SOTA in machine translation.
 - Large Transformer-based models are often pre-trained on large corpora and then fine-tuned for the task at hand.
- Naive application of self-attention to images
 - self-attention only in local neighborhoods for each query pixel instead of globally. (Parmar et al. 2018)
 - Applying the self-attention only local multi-head dot-product self attention blocks can completely replace convolutions. (Hu et al., 2019; Ramachandran et al., 2019; Zhao et al., 2020)
 - employing scalable approximations to global self-attention in order to be applicable to images. (Child et al., 2019)
 - Applying it in blocks of varying sizes, in the extreme case only along individual axes. (Weissenborn et al., 2019)
 - Those are good, but require **complex engineering** to be implemented efficiently on hardware accelerators.
- Attempt to combining CNNs
 - by augmenting feature maps for image classification (Bello et al., 2019)
 - by further processing the output of a CNN using self-attention, e.g. for object detection (Hu et al., 2018; Carion et al., 2020)

Vision Transformer (ViT)



Method

Vision Transformer (ViT)

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \mathbf{x}_p^2 \mathbf{E}; \dots; \mathbf{x}_p^N \mathbf{E}] + \mathbf{E}_{\text{pos}}, \quad \mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}, \mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1) \times D} \quad (1)$$

$$\mathbf{z}'_\ell = \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad \ell = 1 \dots L \quad (2)$$

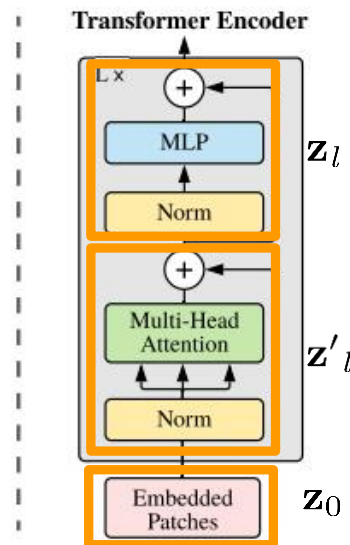
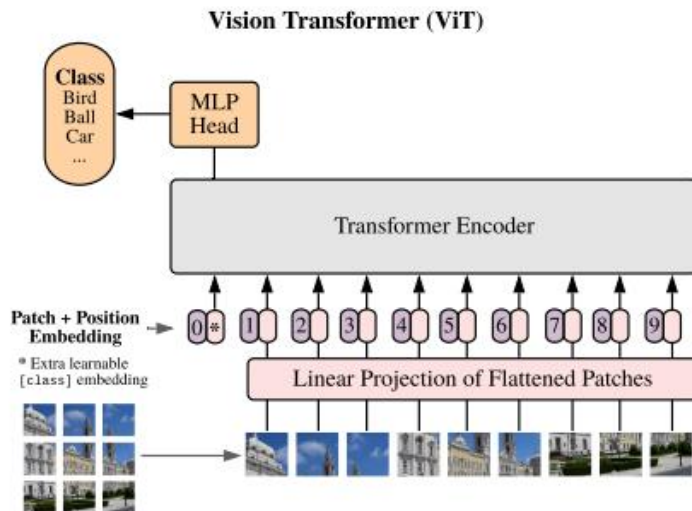
$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, \quad \ell = 1 \dots L \quad (3)$$

$$\mathbf{y} = \text{LN}(\mathbf{z}_L^0) \quad (4)$$

$$\mathbf{x} \in \mathbb{R}^{H \times W \times C}$$

$$\mathbf{x}_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$$

- C = the number of channels
- P = the resolution of patch
- N = the number of patches (HW/P^2)
- D = constant latent vector size
- $\mathbf{y}$ = image representation



Method

Inductive bias

- The inductive bias (also known as learning bias) of a learning algorithm is the **set of assumptions** that the learner uses to **predict outputs of given inputs that it has not encountered**. (thank you Wikipedia!)
- Vision Transformer has much less image-specific **inductive bias** than CNNs.
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- In CNNs, locality, two-dimensional neighborhood structure, and translation equivariance are baked into each layer throughout the whole model.
- In ViT, only MLP layers are local and translationally equivariant, while the self-attention layers are global.

Method

Fine-tuning and higher resolution

- Typically, we pre-train ViT on large datasets, and fine-tune to (smaller) downstream tasks.
- For this, we remove the pre-trained prediction head and attach a zero-initialized $D \times K$ feedforward layer, where K is the number of downstream classes.
- When feeding images of higher resolution, we keep the patch size the same, which results in a larger effective sequence length.
- The ViT can handle arbitrary sequence lengths (up to memory constraints).
- however, the pre-trained position embeddings may no longer be meaningful. We therefore perform 2D interpolation of the pre-trained position embeddings, according to their location in the original image.

Experiments

Setup

- Evaluation capability of ResNet, ViT, hybrid ViT
- Models are Pre-trained on datasets of varying size and evaluate many benchmark tasks
- Used datasets are all de-duplicated.

Dataset	Class	# of images
ILSVRC-2012 ImageNet	1k	1.3M
ImageNet-21k	21k	14M
JFT	18k	303M
ImageNet		
CIFAR-10/100		
Oxford-IIIT Pets	For benchmark tasks	
Oxford Flowers-102		
VTAB		

Experiments

Model Variants.

Model Type	Description
ResNet (BiT)	- Batch Normalization → Group Normalization - Standardized Convolutions
Hybrid (ResNet + ViT)	Intermediate feature maps from ResNet are fed into ViT with patch size = 1
Hybrid Option (i)	Use the output of stage 4 from a standard ResNet50
Hybrid Option (ii)	Remove stage 4 and add the same number of layers to stage 3 (keeping the total number of layers) → Use output of extended stage 3 → Results in 4× longer sequence length and a more expensive ViT model

Model	Layers	Hidden size (D)	MLP Size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

Experiments

Training & Fine-tuning

- We train all models, including ResNets.

Setting	Configuration
Optimizer (pre-training)	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Batch size (pre-training)	4096
Weight decay	0.1 (high, beneficial for transfer)
Learning rate schedule	Linear warmup and decay
Optimizer (fine-tuning)	SGD with momentum
Batch size (fine-tuning)	512
Fine-tuning resolution	512 (ViT-L/16), 518 (ViT-H/14)
Averaging	Polyak averaging, factor = 0.9999

Experiments

Comparison to SOTA

	Ours-JFT (ViT-H/14)	Ours-JFT (ViT-L/16)	Ours-I21k (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	88.55 ± 0.04	87.76 ± 0.03	85.30 ± 0.02	87.54 ± 0.02	88.4/88.5*
ImageNet ReaL	90.72 ± 0.05	90.54 ± 0.03	88.62 ± 0.05	90.54	90.55
CIFAR-10	99.50 ± 0.06	99.42 ± 0.03	99.15 ± 0.03	99.37 ± 0.06	—
CIFAR-100	94.55 ± 0.04	93.90 ± 0.05	93.25 ± 0.05	93.51 ± 0.08	—
Oxford-IIIT Pets	97.56 ± 0.03	97.32 ± 0.11	94.67 ± 0.15	96.62 ± 0.23	—
Oxford Flowers-102	99.68 ± 0.02	99.74 ± 0.00	99.61 ± 0.02	99.63 ± 0.03	—
VTAB (19 tasks)	77.63 ± 0.23	76.28 ± 0.46	72.72 ± 0.21	76.29 ± 1.70	—
TPUv3-core-days	2.5k	0.68k	0.23k	9.9k	12.3k

Table 2: Comparison with state of the art on popular image classification benchmarks. We report mean and standard deviation of the accuracies, averaged over three fine-tuning runs. Vision Transformer models pre-trained on the JFT-300M dataset outperform ResNet-based baselines on all datasets, while taking substantially less computational resources to pre-train. ViT pre-trained on the smaller public ImageNet-21k dataset performs well too. *Slightly improved 88.5% result reported in Touvron et al. (2020).

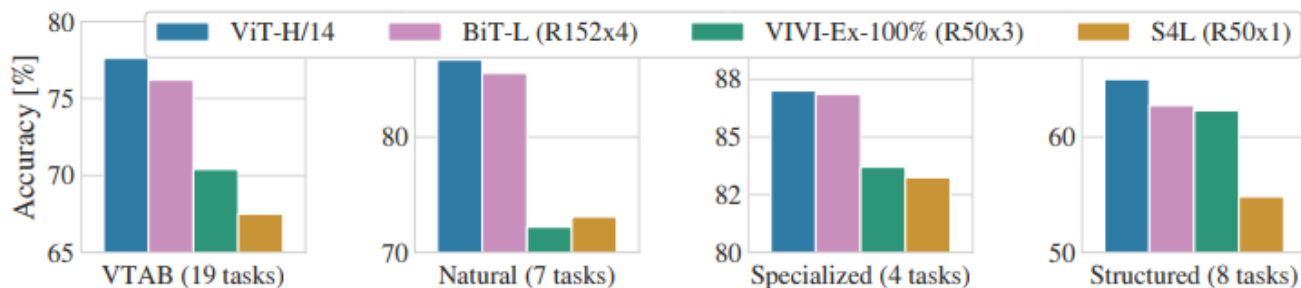


Figure 2: Breakdown of VTAB performance in *Natural*, *Specialized*, and *Structured* task groups.

Experiments

Pre-training data requirements

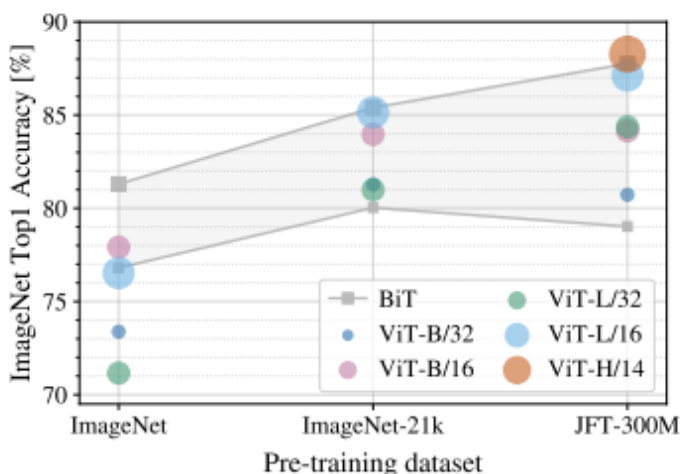


Figure 3: Transfer to ImageNet. While large ViT models perform worse than BiT ResNets (shaded area) when pre-trained on small datasets, they shine when pre-trained on larger datasets. Similarly, larger ViT variants overtake smaller ones as the dataset grows.

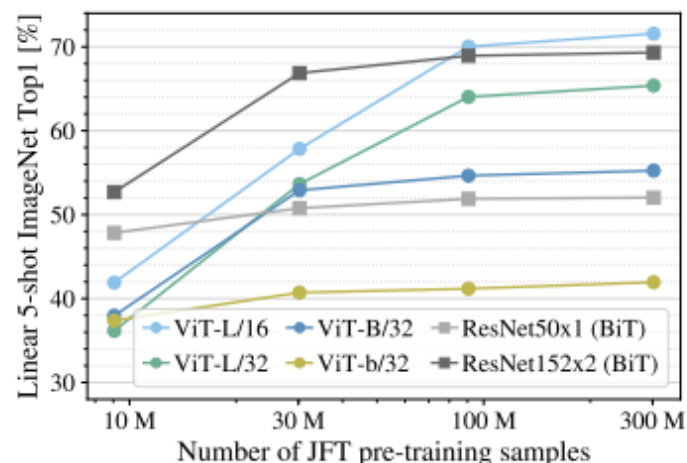


Figure 4: Linear few-shot evaluation on ImageNet versus pre-training size. ResNets perform better with smaller pre-training datasets but plateau sooner than ViT, which performs better with larger pre-training. ViT-b is ViT-B with all hidden dimensions halved.

Experiments

Pre-training data requirements

Model Type	Models	Pre-training Epochs	Notes / Observations
ResNets	R50x1, R50x2, R101x1, R152x1, R152x2, R200x3	7 (most), 14 (R152x2, R200x3)	R152x2 and R200x3 trained longer
Vision Transformers	ViT-B/32, ViT-B/16, ViT-L/32, ViT-L/16, ViT-L/16, ViT-H/14	7 (most), 14 (L/16, H/14)	Dominates ResNets in performance/compute trade-off; 2–4× less compute needed
Hybrids (ResNet+ViT)	R50+ViT-B/32, R50+ViT-B/16, R50+ViT-L/32, R50+ViT-L/16, R50+ViT-L/16	7 (most), 14 (R50+ViT-L/16)	Slightly better than ViT at small compute budgets; difference disappears for larger models

Experiments

Inspecting ViT

- We train all models, including ResNets.

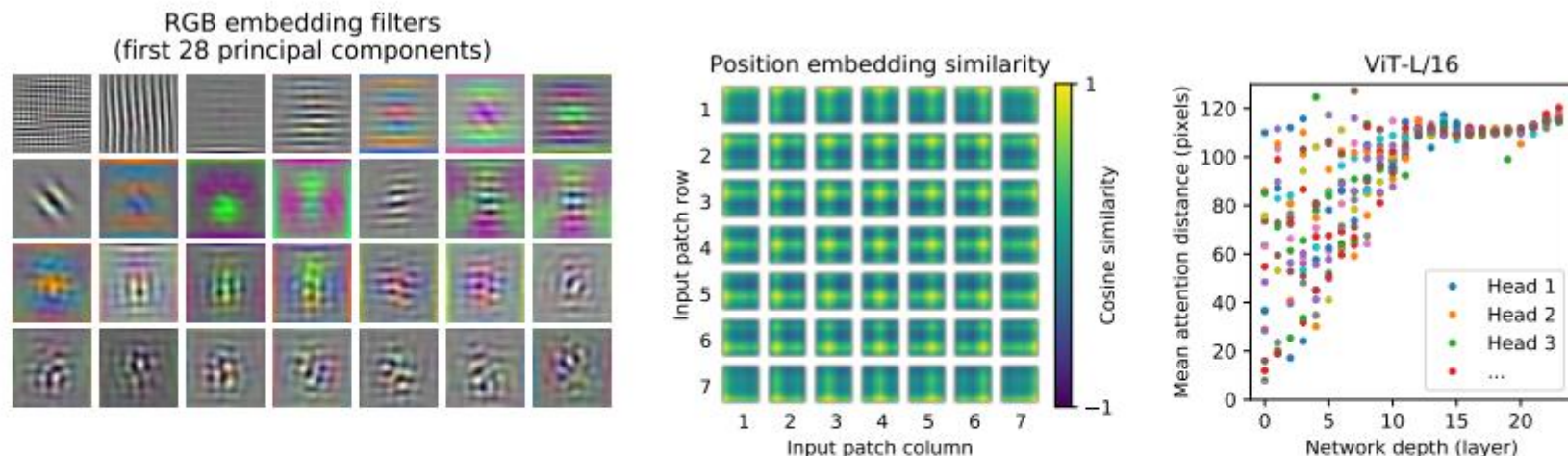


Figure 7: **Left:** Filters of the initial linear embedding of RGB values of ViT-L/32. **Center:** Similarity of position embeddings of ViT-L/32. Tiles show the cosine similarity between the position embedding of the patch with the indicated row and column and the position embeddings of all other patches. **Right:** Size of attended area by head and network depth. Each dot shows the mean attention distance across images for one of 16 heads at one layer. See Appendix D.7 for details.

Conclusion

ViT

- Direct application of Transformers to images and treat images as sequences of patches
- No image-specific biases beyond patch extraction and uses standard Transformer encoder
- Pre-training on large datasets enables strong performance
- Achieves state-of-the-art results on multiple image classification benchmarks
- Challenges include extending to detection and segmentation, improving self-supervised pre-training, and exploring larger-scale models